

智能阶梯轴自动上料与AI视觉检测系统项目书

60x20 mm工件 | 0.12 kg | 12件/分钟

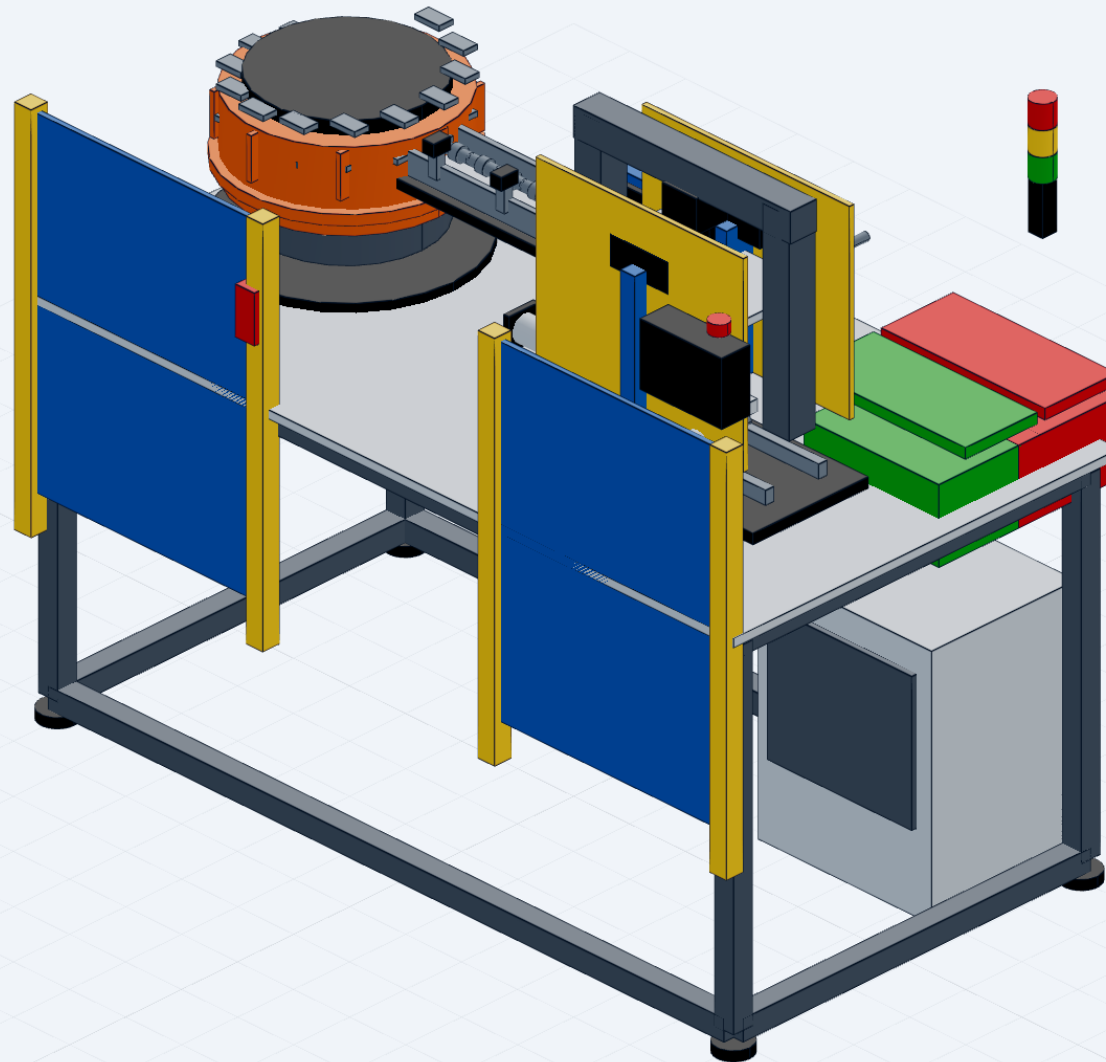
振动盘定向、直线送料、双气缸分料、V形定位、视觉/激光检测和OK/NG分流完整工作站。

离散螺旋轨道与可调直线送料
5 s循环动作分解
视觉轮廓与激光测径组合
阀岛、电控、安全门和防护一体化

Editable CAD source + STEP + GLB + calculations + BOM + manufacturing plan + validation evidence

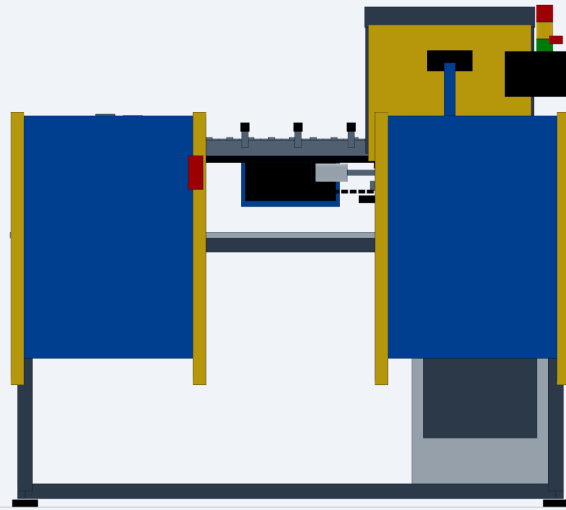
01 Detailed assembly overview

ISO

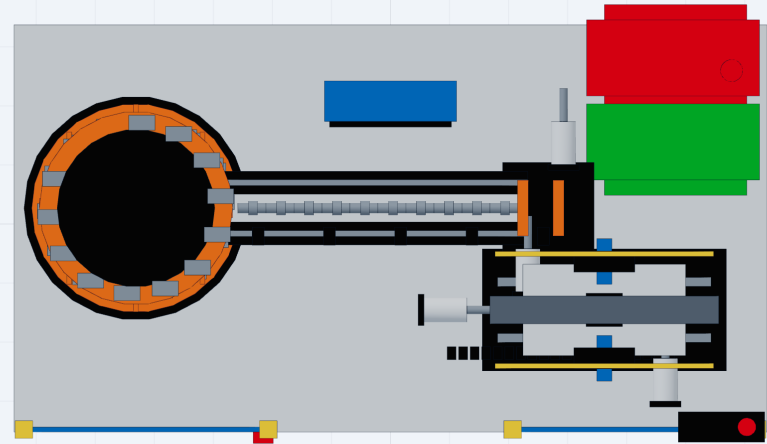


02 Orthographic and opposite views

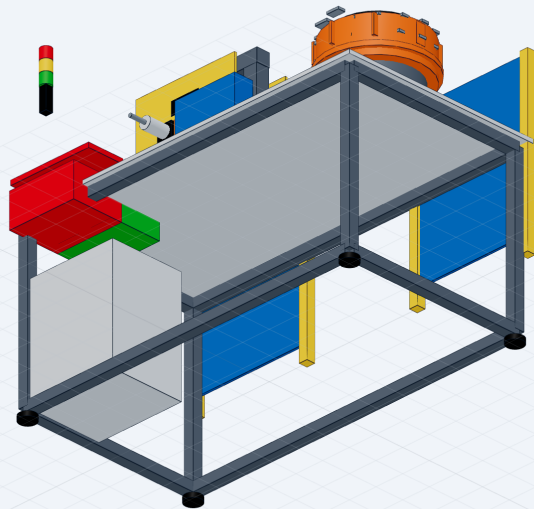
FRONT



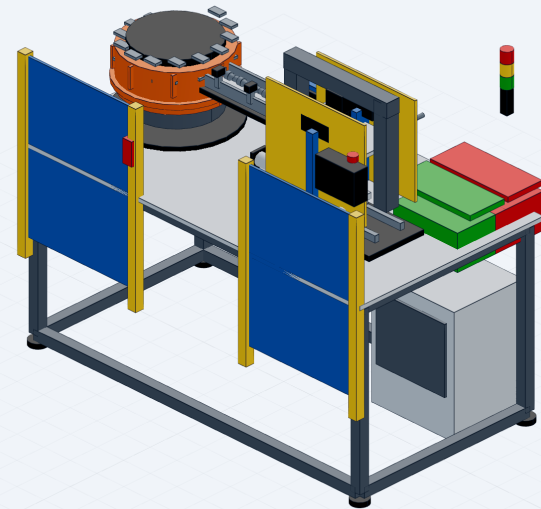
TOP



ISO



ISO



03 Parameters and preliminary calculations

parameter	value	unit	status
workpiece_length	60	mm	design baseline
workpiece_max_diameter	20	mm	design baseline
workpiece_mass	0.12	kg	design baseline
target_rate	12	parts/min	target
cycle_time	5	s	calculated
air_pressure	0.5	MPa	assumed
transfer_cylinder_bore	20	mm	selected baseline
separator_cylinder_bore	16	mm	selected baseline
model_bounding_box	1480x862.5x1324.5	mm	verified
step_faces	924	count	verified
step_edges	1698	count	verified

One 5 s cycle is allocated as: confirmation 0.4 s, separation 0.7 s, transfer 1.0 s, settle 0.4 s, image/laser acquisition 1.0 s, judgement 0.3 s, eject/return 1.2 s.

For a 0.12 kg part on a guide with assumed friction coefficient 0.2 and 0.5 m/s² acceleration, required force with factor 3 is below 1 N. A 16 mm bore cylinder at 0.5 MPa provides about 100 N theoretical force; a 20 mm transfer cylinder provides about 157 N. This leaves margin for seals, guide friction and occasional misalignment.

Five photoelectric checks cover bowl exit, track queue, first gate, second gate and V-block arrival. Laser micrometer measures the critical diameter; the top camera checks overall contour, step position, orientation and surface-defect candidates.

04 BOM, manufacturing and validation

item	component	qty	specification
1	vibratory bowl feeder	1	400 mm class
2	linear feeder track	1	620 mm adjustable rails
3	separator cylinders	2	16 mm bore / 65 mm stroke
4	transfer cylinder	1	20 mm bore / 170 mm stroke
5	eject cylinder	1	16 mm bore / 85 mm stroke
6	linear guides	2	420 mm
7	V-block fixture	1	hardened replaceable inserts
8	photoelectric sensors	5	24 VDC
9	industrial camera	1	top view
10	telecentric lens	1	matched field of view
11	backlight	1	320 x 180 mm
12	laser micrometer pair	1	emitter/receiver
13	pneumatic valve island	1	8-station
14	PLC/HMI/electrical cabinet	1 set	24 VDC control
15	OK and NG bins	2	removable
16	aluminium-profile frame	1 set	40 series
17	polycarbonate guarding	1 set	door interlock
18	cable chain	1	transfer axis

Manufacturing / assembly

Track and fixture surfaces are CNC milled with replaceable wear strips. V-block inserts are tool steel, heat treated and ground. Frame members are cut aluminium profiles with dowel-located station plates. Sensor brackets use slots for calibration. Guard panels are polycarbonate in aluminium frames. Pneumatic tubing and electrical cables are separated and labelled for maintenance.

Validation boundary

自动上料与视觉检测系统

设计验证补充说明（2026-07-24更新）

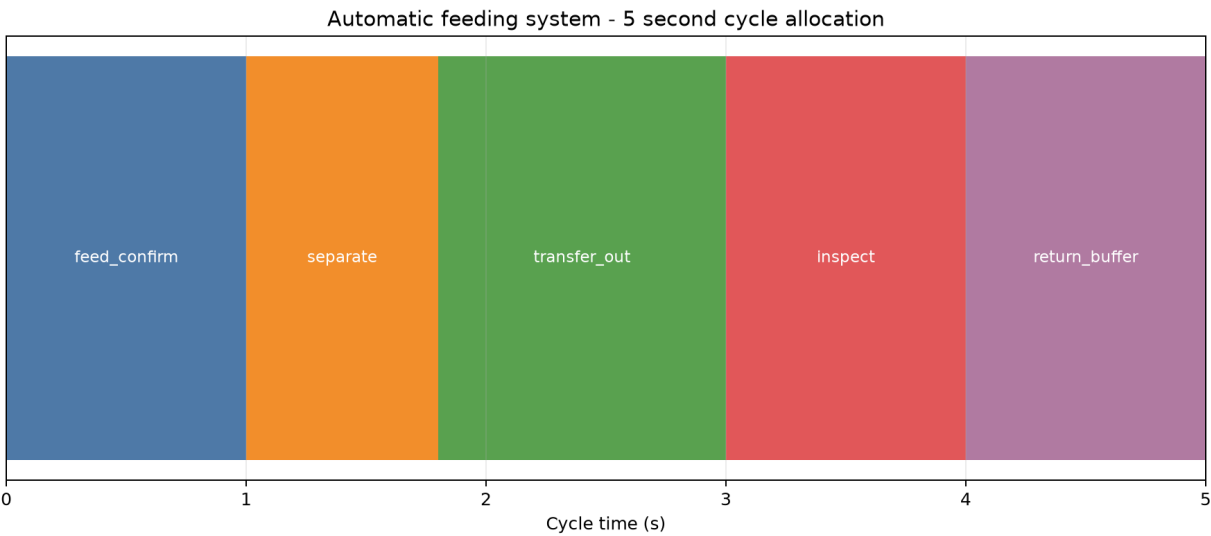
本次更新

针对上一版底板挠度超限完成加厚、模型更新和复算闭环。

设计节拍	5 s/循环
底板优化	12 mm → 14 mm
优化后挠度	0.611 mm
挠度限值	0.80 mm

结构复核结果

分析方法	一维 Euler-Bernoulli 梁有限元
材料	6061-T6 铝合金
等效截面	70 × 14 mm
最大弯曲应力	9.84 MPa
最大挠度	0.611 mm
屈服安全系数	24.39
结论	初步通过



实体交叠筛查与整改状态

详细STEP实体数	179
包围盒候选对	233
精确共同体积对	227
求交失败数	0

判断说明

精确共同体积不等同于全部都是设计错误，其中可能包含轴孔配合、紧固件穿入、焊接/并体构造以及真实零件穿透。仓库已按P1/P2/P3生成整改工作表，记录空间区域推断、几何类别、处理动作和关闭状态。未完成逐零件命名和关闭前，不声明整机为无干涉制造装配。

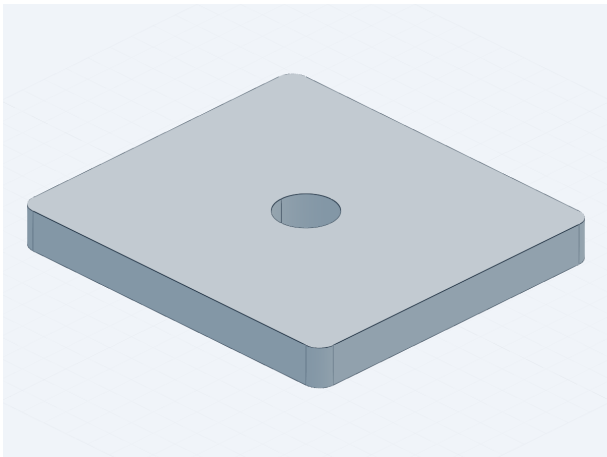
原生格式状态

已验证交付包括基础SolidWorks 2023零件、增强STEP、可编辑DXF、A3 PDF、参数化重建源码、运动/节拍验证和结构复核源码。NX 12原生特征建模受本机许可证错误-10005限制；原生受约束装配和三维实体FEA尚未伪造为已完成，仓库保留失败证据与恢复步骤。

适用边界：结构结果用于方案阶段整体刚度和名义应力初筛，不能代替孔边、焊缝、轴承座、螺栓预紧、接触、屈曲、疲劳和冲击工况的三维分析。节拍、精度、视觉准确率和连续运行能力需通过最终控制系统及实体样机验证。

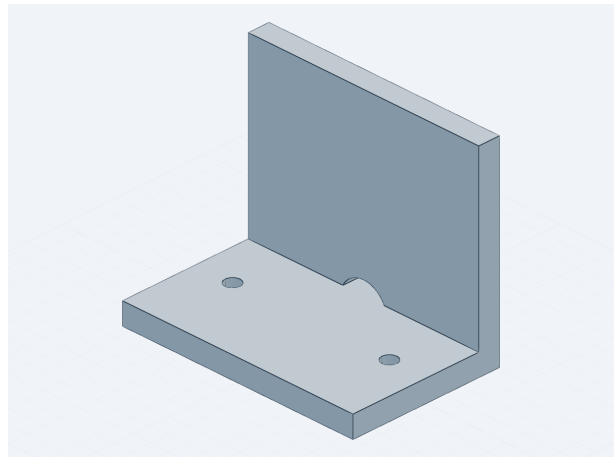
第二批生产型关键零件

每个零件包含单实体STEP、毫米制DXF、A3 PDF、BOM、截图及参数化生成源码。



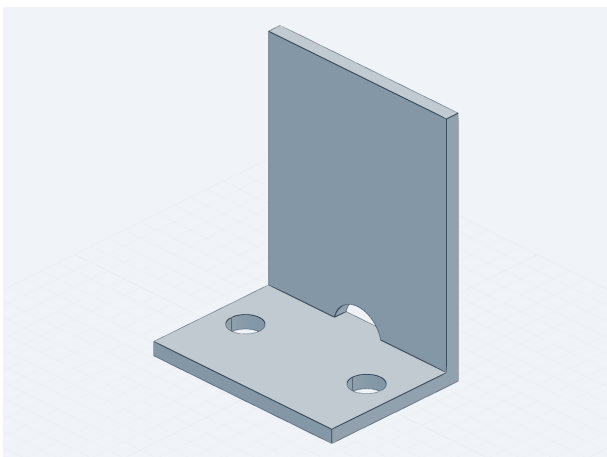
FI-005 separator_gate

45 steel | Qty 2 | 110x100x12 mm



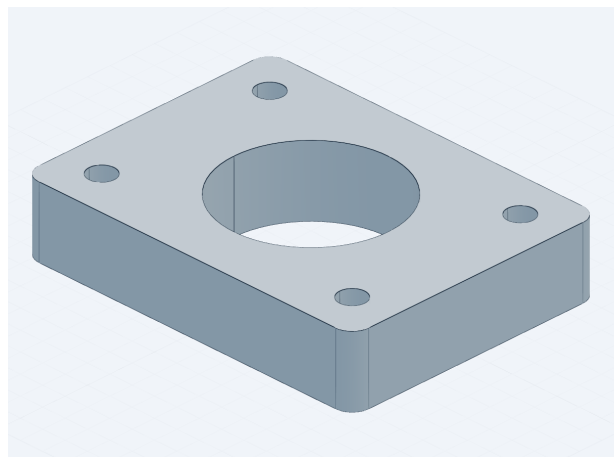
FI-006 hard_stop_bracket

Q235B | Qty 1 | 110x70x90 mm



FI-007 photo_sensor_bracket

SUS304 | Qty 5 | 45x32x55 mm

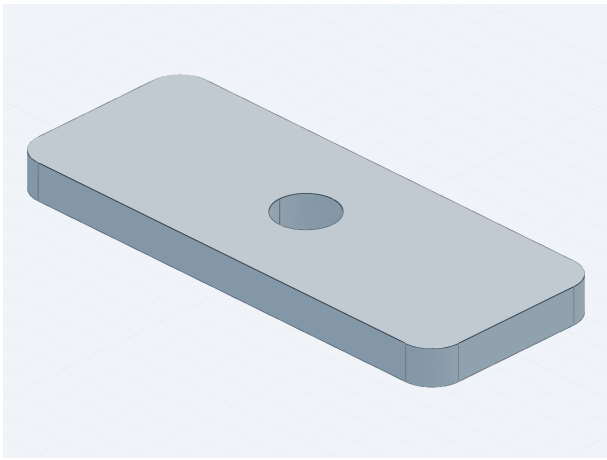


FI-008 camera_crossbeam_mount

6061-T6 | Qty 1 | 120x90x24 mm

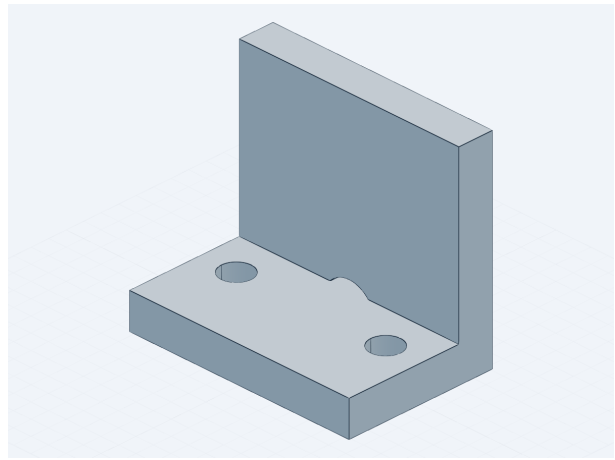
第三批连接与安装零件

每个零件包含单实体STEP、毫米制DXF、A3 PDF、BOM、截图及参数化生成源码。



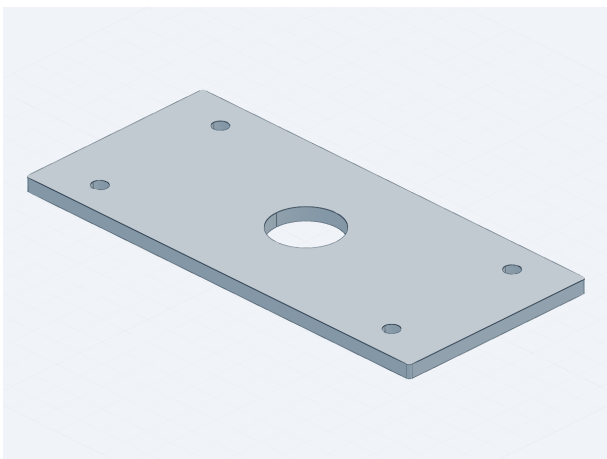
FI-009 guide_rail_clamp_plate

SUS304 | Qty 8 | 80x32x6 mm



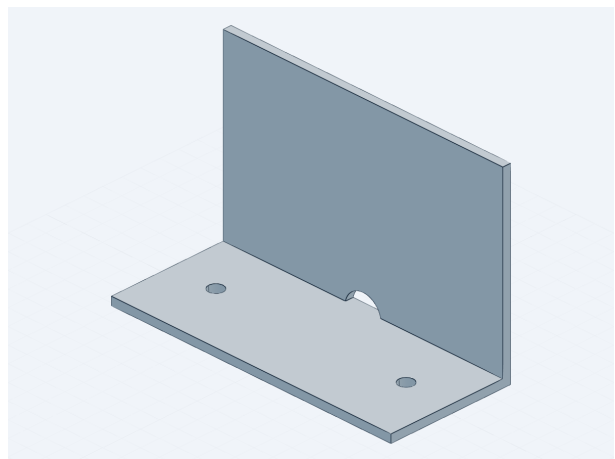
FI-010 cylinder_rod_clevis

45 steel | Qty 4 | 52x34x46 mm



FI-011 valve_manifold_mounting_plate

Q235B | Qty 1 | 260x120x8 mm



FI-012 backlight_mount_bracket

5052-H32 | Qty 2 | 140x60x90 mm